

|      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Que. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 3  | 2  | 2  | 3  | 2  | 3  | 1  | 2  | 3  | 3  | 1  | 4  | 1  | 4  | 2  | 2  | 4  | 1  | 1  | 4  |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | 1  | 3  | 2  | 1  | 3  | 4  | 3  | 2  | 2  | 3  | 4  | 4  | 2  | 4  | 4  | 2  | 2  | 4  | 4  | 3  |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| Ans. | 2  | 1  | 3  | 4  | 1  | 3  | 4  | 2  | 3  | 1  | 3  | 1  | 4  | 3  | 1  | 1  | 2  | 3  | 4  | 2  |
| Que. | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| Ans. | 4  | 1  | 1  | 3  | 3  | 1  | 4  | 2  | 1  | 1  | 1  | 1  | 1  | 3  | 4  | 3  | 2  | 2  | 3  | 2  |
| Que. | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |    |    |    |    |    |    |    |    |    |    |
| Ans. | 3  | 2  | 4  | 4  | 2  | 3  | 2  | 4  | 4  | 3  |    |    |    |    |    |    |    |    |    |    |

# JEE (Main + Advanced) : LEADER COURSE PHASE- III to VII (SCORE-I)

Test Type : PART TEST

Test Pattern : JEE-Advanced

**TEST DATE : 05 - 03 - 2017**

## PAPER-1

### PART-1 : PHYSICS

### SOLUTION

#### SECTION-I

1. Ans. (A,B,D)

Sol. Perimeter is decreasing at a rate of  $2v$

$$\therefore \frac{d}{dt}(2\pi r) = 2v \Rightarrow \frac{dr}{dt} = \frac{v}{\pi}$$

$$\therefore r = (r_0 - \frac{v}{\pi} t)$$

$$\phi = B \cdot \pi r^2 \Rightarrow \varepsilon = \left| \frac{-d\phi}{dt} \right| = B \cdot 2\pi \cdot r \frac{dr}{dt}$$

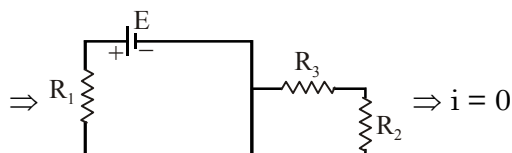
$$\therefore \varepsilon = 2B\pi(r_0 - \frac{v}{\pi} t) \cdot \frac{v}{\pi} = 2Bv(r_0 - \frac{v}{\pi} t)$$

$$I = \frac{\phi}{R} = \frac{2Bvr}{\lambda \cdot 2\pi r} = \frac{Bv}{\pi\lambda}$$

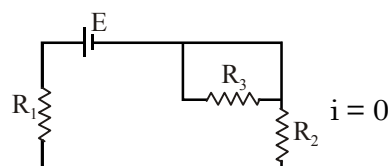
2. Ans. (A,B)

Sol. Just after switch is closed, L is open circuit and C is short circuit

Just after



Long time



3. Ans. (A,B,D)

Sol.  $p_0 + h_1\rho_1g = p_0 + h_2\rho_1g + h_3\rho_3g$

4. Ans. (B,D)

Sol.  $V_{RL} = i_0 \times \sqrt{\omega^2 L^2 + R^2}$

$$= \frac{V_0}{\sqrt{\left(\omega L - \frac{1}{\omega C}\right)^2 + R^2}} \times \sqrt{\omega^2 L^2 + R^2} \quad \omega = 0,$$

$$V_{RL} \propto V_0 \times \omega C$$

$$V_C = i_0 \times \frac{1}{\omega C}$$

$$\omega = \infty$$

$$V_{RL} = V_0$$

$$= \frac{V_0}{\sqrt{(\omega^2 LC - 1)^2 + \omega^2 R^2 C^2}}$$

$$\omega = 0$$

$$V_C = \frac{V_0}{\omega} \sqrt{LC}$$

$$\omega = \infty$$

5. Ans. (B,D)

Sol. Applying ampere's law

$$B 2\pi r = \mu_0 i_{\text{enclosed}}$$

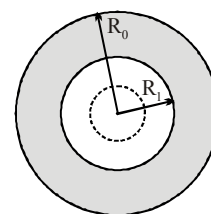
$$\text{Option (B)} \quad i_{\text{enclosed}} = 0$$

$$\Rightarrow B = 0$$

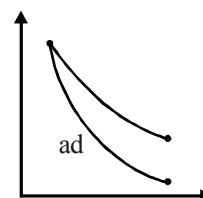
$$\text{Option (D)} \quad i_{\text{enclosed}} = 0$$

6. Ans. (A,D)

7. Ans. (A,B)



Sol.

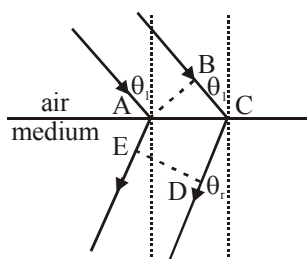


8. **Ans. (B,D)**

**Sol.** The geometrical construction shown in figure is important for developing the mathematical description of interference. It is subject to misinterpretation, however, as it might suggest that the interference can only occur at the position of the screen. A better diagram for this situation is figure, which shows paths of destructive and constructive interference all the way from the slits to the screen. These paths would be made visible by the smoke. Since fringes are present everywhere in space they are non-localised and their shapes are hyperboloid.

9. **Ans. (A)**

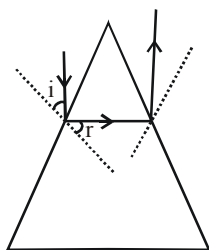
**Sol.**  $\mu d_{AE} = d_{BC} + \mu d_{CD}$   
 $\mu(d_{AE} - d_{CD}) = d_{BC}$



$$\mu = \frac{C}{v} = -\sqrt{\epsilon_r \mu_r}$$

$$(d_{AE} - d_{CD}) = -\frac{d_{BC}}{\sqrt{\epsilon_r \mu_r}}$$

10. **Ans. (C)**



**Sol.**

$$\frac{\sin i}{\sin r} = \mu$$

$$\mu_e < 0, \epsilon_r < 0$$

so refractive index is negative,  
so final wave return to it's  
same side where it is incident.

11. **Ans. (D)**

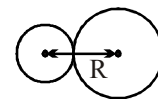
**Sol.** Fring width;  $\beta = \frac{D\lambda}{d}$

If  $V$  decreases, then  $\lambda$  increases & hence  $\beta$  increases.

12. **Ans. (A)**

13. **Ans. (D)**

**Sol.**  $d = R_N + R_p$



$$R = R_0 A^{1/3} = R_0 14^{1/3} + R_0 4^{1/3}$$

14. **Ans. (C)**

$$\text{Sol. } \frac{1}{2} \mu v_{\text{rel}}^2 = \frac{k q_1 q_2}{d}$$

$$\frac{1}{2} \times \frac{4m \times 14m}{4m + 14m} v_{\text{rel}}^2 = \frac{k \times 2 \times e \times 7e}{R_0(14^{1/3} + 2^{2/3})}$$

$$\frac{1}{2} \times 4m v_{\text{rel}}^2 = \frac{18ke^2}{R_0(14^{1/3} + 2^{2/3})}$$

## SECTION-II

1. **Ans. (A) → (P,Q,R,S); (B) → (P,Q,R,S); (C) → (P,Q,R,S,T); (D) → (R,S)**

## SECTION-IV

1. **Ans. 3**

2. **Ans. 8**

**Sol.** Process A → B

$$W_{AB} = \int P dv = \int \frac{3}{2} T^{1/2} dv$$

$$= \int \frac{3}{2} T^{1/2} \times \frac{1}{3} RT^{-1/2} dT$$

$$\text{On solving, } W_{AB} = 50 R = 50 \times 8.3 = 415 \text{ J}$$

Process B → C

$$U = \frac{1}{2} V^{1/2}$$

$$\frac{3}{2} RT = \frac{1}{2} V^{1/2} \Rightarrow 3PV^{1/2} = 1$$

$$\therefore P = \frac{1}{3\sqrt{V}}$$

$$\text{Now } W_{BC} = \int P dv = \int_{100}^{1600} \frac{1}{3\sqrt{V}} dv = \frac{2}{3} \sqrt{V}$$

$$= \frac{2}{3} [40 - 10] = \frac{2}{3} \times 30 = 20 \text{ J}$$

$$\text{Total } W = 415 + 20 = 435$$

3. **Ans. 2**

**Sol.**  $P_{ab} = \frac{\text{heat energy}}{\text{time}}$

$$(\text{Intensity}) (\text{cross section area}) = \frac{ms\Delta T}{t}$$

$$t = \frac{(4 \times 10^{-3})(4200)(5)}{(1000)(1.4 \times 10^{-4})} = 600 \text{ s} = 10 \text{ min.}$$

4. **Ans. 8**

**Sol.** Energy available =  $\frac{1}{2} \mu v_{\text{rel}}^2 = Q$  value.

$$= \frac{1}{2} \times \frac{7 \times 1}{7+1} \times v_{\text{rel}}^2 = Q \text{ value.}$$

$$\Rightarrow \frac{1}{2} \times v_{\text{rel}}^2 = Q \times \frac{8}{7}$$

$$K_1 = 1645 \times \frac{8}{7} = 1880 \text{ keV}$$

5. **Ans. 2**

**Sol.**  $\lambda_{\text{th}} = \frac{hc}{eV_a}$

$$\frac{1}{\lambda_{K\alpha}} = R(z-1)^2 \left( \frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$\frac{13}{10} (\lambda_{K\alpha} - \lambda_{\text{th}}) = \left( \lambda_{K\alpha} - \frac{\lambda_{\text{th}}}{2} \right)$$

$$\frac{3}{10} \lambda_{K\alpha} = \left( \frac{13}{10} - \frac{1}{2} \right) \lambda_{\text{th}}$$

$$\frac{3}{10} \left( \frac{4 \times 10^{-7}}{3(z-1)^2} \right) = \left( \frac{8}{10} \right) \frac{12.4 \times 10^{-7}}{15.5 \times 10^3}$$

$$\Rightarrow \frac{5000}{8} = (z-1)^2$$

$$625 = (z-1)^2 \Rightarrow z = 26$$

## PART-2 : CHEMISTRY

## SOLUTION

### SECTION - I

1. **Ans. (A,C,D)**

2. **Ans. (B,C)**

$$\frac{dN}{N} \propto \sqrt{\frac{M}{T}} du$$

3. **Ans. (A, C, D)**

4. **Ans. (A, C)**

5. **Ans. (A, B)**

6. **Ans. (A,B,C,D)**

7. **Ans. (B)**

8. **Ans. (B,C,D)**

9. **Ans. (A)**

$$\Delta G^\circ = -nFE^\circ$$

$$E^\circ_{298K} = 0.07 \text{ volt}$$

$$= - \frac{(1)(96500)(0.07)}{1000}$$

$$= -96.5 \times 0.07 = -6.755$$

10. **Ans. (C)**

$$\Delta S^\circ = -nF \frac{dE^\circ}{dT}$$

$$\left( \frac{dE^\circ}{dT} \right)_{298K} = 5 \times 10^{-4} - 7 \times 10^{-6} \left( \frac{T}{K} - 298 \right)$$

$$= 5 \times 10^{-4} - 7 \times 10^{-6} (0)$$

$$= 5 \times 10^{-4}$$

$$\Delta S^\circ = + (1) (96500) (5 \times 10^{-4})$$

$$= + 482500 \times 10^{-4}$$

$$= +48.25 \text{ JK}^{-1}\text{Mol}^{-1}$$

11. **Ans. (D)**

12. **Ans. (D)**

13. **Ans. (C)**

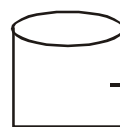
14. **Ans. (D)**

### SECTION - II

1. **Ans. (A)→(Q,R); (B)→(P,S); (C)→(P,R,T); (D)→(Q,T)**

### SECTION - IV

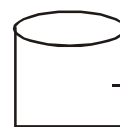
1. **Ans. 5**



$$[\text{NaCl}] = 0.025 \text{ M} = C_1$$

$$V_1 = 2V$$

$$T_1 = 273 \text{ K}$$



$$[\text{Na}_2\text{SO}_4] = 0.1 \text{ M}$$

$$V_2 = 3V$$

$$T_2 = 273 \text{ K}$$

$$\Pi = \left[ \frac{(c_1 v_1 i_1) + (c_2 v_2 i_2)}{v_1 + v_2} \right] 22.4$$

$$= \left[ \frac{(0.025 \times 2v \times 2) + (0.1 \times 3v \times 3)}{5v} \right] 22.4$$

$$= \left( \frac{0.1v + 0.9v}{5v} \right) 22.4 = \frac{22.4}{5}$$

$$\Pi = 4.48$$

2. **Ans. 2**

3. **Ans. 6**

4. **Ans. 4**

5. **Ans. 7**

**PART-3 : MATHEMATICS**

**SOLUTION**

**SECTION-I**

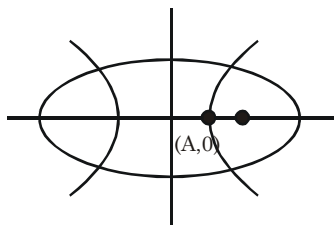
1. **Ans. (B,C,D)**

$$y = m_1x - 2m_1 - m_1^3$$

$$6 = m_1 - m_1^3 \Rightarrow m_1 = -2 \Rightarrow m = \frac{1}{2}$$

$$P(4,4) \Rightarrow r = \sqrt{5}$$

2. **Ans. (A,B,C)**



Hyperbola and ellipse will be confocal with focus  $(\pm 2\sqrt{2}, 0)$

$$1 - e_1^2 = \frac{1}{9} \Rightarrow e_1 = \frac{2\sqrt{2}}{3} \Rightarrow e_2 = \frac{3}{2\sqrt{2}}$$

$$\therefore \frac{x^2}{A^2} - \frac{y^2}{B^2} = 1 \Rightarrow A = \frac{8}{3}$$

$$B^2 = \frac{64}{9} \left( \frac{9}{8} - 1 \right) \Rightarrow B^2 = \frac{8}{9}$$

$$\frac{9x^2}{64} - \frac{9y^2}{8} = 1$$

3. **Ans. (A,B,C,D)**

$$\begin{vmatrix} 2\alpha & 1 & 1 \\ \alpha & 1 & 2 \\ \beta^2 & 1 & -3\alpha \end{vmatrix} = 0$$

$$\Rightarrow 2\alpha(-3\alpha - 2) - \alpha(-3\alpha - 1) + \beta^2 = 0$$

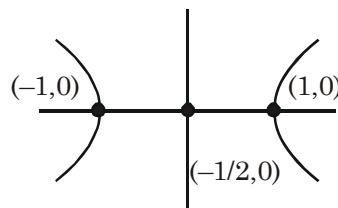
$$\Rightarrow -3\alpha^2 + \beta^2 - 3\alpha = 0$$

$$\Rightarrow 3(\alpha^2 + \alpha) - \beta^2 = 0$$

$$\Rightarrow \left( \alpha + \frac{1}{2} \right)^2 - \frac{\beta^2}{3} = \frac{1}{4}$$

$$\text{so conic is } \left( x + \frac{1}{2} \right)^2 - \frac{y^2}{3} = \frac{1}{4}$$

$$\text{whose center is } \left( -\frac{1}{2}, 0 \right)$$



$$e^2 - 1 = \frac{3}{1} \Rightarrow e = 2$$

4. **Ans. (A,C)**

$$\frac{2k_1\pi}{16} = \frac{2k_2\pi}{72} \Rightarrow k_1 = \frac{2}{9}k_2$$

$$k_2 = 0, 9, \dots, 63 \rightarrow 8 \text{ values}$$

$$\text{so } n(A \cap B) = 8.$$

$$e^{\left( \frac{2k_1\pi}{16} + \frac{2k_2\pi}{72} \right)i} = \left\{ \frac{\pi/4}{\pi/36} = 9 \right\}.$$

as 1<sup>st</sup>, 10<sup>th</sup>, 19<sup>th</sup> roots are common.

$\therefore k_2 = 0, 1, \dots, 8$  and  $k_1 = 0, 1, \dots, 16$ .

will give all possible  $z_1 z_2 \rightarrow 144$

5. **Ans. (B)**

$$A = (d_1, d_2, d_3, d_4)$$

$$A^4 = (d_1^4, d_2^4, d_3^4, d_4^4) = I$$

$$\Rightarrow d_1^4 = d_2^4 = d_3^4 = d_4^4 = I$$

$\Rightarrow d_1, d_2, d_3, d_4$  are fourth roots of unity  
 $d_1 + d_2 + d_3 + d_4 = 0$

$$\Rightarrow \left( 2 \cdot \frac{4!}{2!2!} \right) + 4! = 36 \text{ ways are there to}$$

assign values to  $d_1, d_2, d_3, d_4$ .

Also  $d_1 d_2 d_3 d_4$  is product of 4<sup>th</sup> roots of unity which is -1 or 1

when 1, -1, 1, -1 or i, -i, i, -i are used.

6. **Ans. (A,B)**

$$\text{Let } A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\Rightarrow AB = BA$$

$$\Rightarrow \begin{bmatrix} a & a+b & a+b+c \\ d & d+e & d+e+f \\ g & g+h & g+h+i \end{bmatrix} = \begin{bmatrix} a+d+g & b+e+h & c+f+i \\ d+g & e+h & f+i \\ g & h & i \end{bmatrix}$$

$$g = 0, d = h = 0, a = e = i, b = f$$

$$\Rightarrow A = \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}$$

- (A)  $a = b = c = 1 \Rightarrow |A| = 1$   
 (B)  $\text{trace}(A) = 6 \Rightarrow a = z \Rightarrow |A| = 8$   
 (C)  $|A| = a^3$  which may or may not be zero  
 (D)  $3a + 2b + c = 6$

| a | b          | c |                          |
|---|------------|---|--------------------------|
| 0 | 0, 1, 2, 3 |   | 4 matrices               |
| 1 | 0, 1       |   | $\rightarrow$ 2 matrices |
| 2 | 0          | 0 | 1 matrix                 |

7. **Ans. (A,C)**

$\vec{C} = \vec{A} \times \vec{B} \Rightarrow \vec{A} \cdot \vec{C} = 0 \Rightarrow 2a + 2b - c = 0$   
 for (A) and (B)  $a + b + c = 0 \Rightarrow c = 0, a + b = 0$

$$\text{as } a^2 + b^2 + c^2 = 1 \Rightarrow \begin{cases} \left( \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right) \\ \left( -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \end{cases}$$

(C) and (D)  $\left. \begin{matrix} a^2 + b^2 + c^2 = 1 \\ 2a + 2b = c \end{matrix} \right\}$  are not satisfied  
 for any triplet of integers.

8. **Ans. (A,B,D)**

If  $n$  coplanar lines are there such that no 3 are concurrent and no two are parallel then the number of parts in which they will divide their plane is  $2 + 2 + 3 \dots n$

$$= \frac{n^2 + n + 2}{2}$$

$(n + 1)^{\text{th}}$  plane will be intersected by previous  $n$  planes in  $\frac{n^2 + n + 2}{2}$  lines

$$\therefore f(n+1) = f(n) + \frac{n^2 + n + 2}{2}$$

$$\Rightarrow f(n) = f(n-1) + \frac{(n-1)^2 + (n-1) + 2}{2}$$

$$= 2 + \frac{1^2 + 1 + 2}{2} + \frac{2^2 + 2 + 2}{2} + \dots + \frac{(n-1)^2 + (n-1) + 2}{2}$$

$$= 2 + \frac{1}{2} \left[ \frac{(n(n-1)(2n-1))}{6} + \frac{n(n-1)}{2} + 2(n-1) \right]$$

$$= \frac{n^3 + 5n + 6}{6}$$

### Paragraph for Question 9 & 10

$$x_1 = \sqrt{5} \Rightarrow V(5, 0)$$

$$y^2 = -4a(x - 5)$$

$$\Rightarrow \frac{x^2}{9} - a(x - 5) = 1$$

$$\Rightarrow x^2 - 9ax + 45a - 9 = 0$$

$$D = 81a^2 - 4(45a - 9) = 0$$

$$= 9a^2 - 4(5a - 1) = 0$$

$$9a^2 - 20a + 4 = 0 \Rightarrow a = 2, \frac{2}{9}$$

at  $a = 2 \Rightarrow x = 9$  rejected

$$a = \frac{2}{9} \Rightarrow x = 1 \Rightarrow A\left(1, \frac{4\sqrt{2}}{3}\right); B\left(1, -\frac{4\sqrt{2}}{3}\right)$$

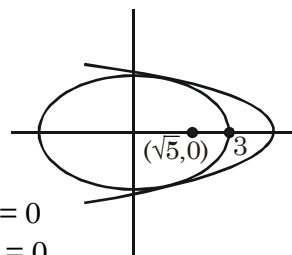
Normal at A

$$\frac{9x}{1} - \frac{4y}{4\sqrt{2}/3} = 5 \Rightarrow 9x - \frac{3}{\sqrt{2}}y = 5$$

$$9x\sqrt{2} - 3y = 5\sqrt{2}$$

$$\text{tangent at A } 3x + 9y\sqrt{2} = 27 \Rightarrow c(9, 0)$$

$$\Rightarrow \Delta ABC = \frac{1}{2} \cdot 8 \cdot \frac{8\sqrt{2}}{3} = \frac{32\sqrt{2}}{3}$$



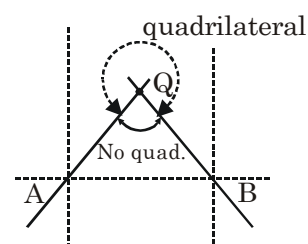
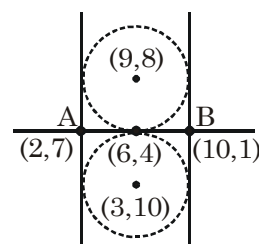
9. **Ans. (C)**

10. **Ans. (B)**

### Paragraph for Question 11 & 12

$a + b = 13$  or  $17$

$L_4$  passes through  $Q(12, 12)$



11. **Ans. (C)**

12. **Ans. (C)**

### Paragraph for Question 13 to 14

$$F(n, P) = \frac{n^P - \binom{n}{1}(n-1)^P + \binom{n}{2}(n-2)^P - \dots + (-1)^{n-1} \binom{n}{n-1} 1^P}{n^P} \dots (i)$$

$$F(4, 6) = \frac{4^6 - \binom{4}{1}3^6 + \binom{4}{2}2^6 - \binom{4}{3}1^6}{4^6}$$

$$= \frac{4096 - 2916 + 384 - 4}{4096} = \frac{1560}{4096} = \frac{195}{512}$$

Also if  $P < n$  then probability is zero and if  $P = n$  by (1)

$$\frac{n^n - \binom{n}{1}(n-1)^n + \dots + (-1)^{n-1}\binom{n}{n-1}1^n}{n^n} = \frac{n!}{n^n}$$

$$\Rightarrow (-1)^{n-1} [{}^nC_1 1^n - {}^nC_2 2^n + {}^nC_3 3^n - \dots + (-1)^{n-1} {}^nC_n n^n] = n!$$

$\therefore$  D is right.

13. Ans. (A)

14. Ans. (D)

### SECTION - II

1. Ans. (A)  $\rightarrow$  (P, Q, R, S, T); (B)  $\rightarrow$  (Q, S); (C)  $\rightarrow$  (P, Q, R, S, T); (D)  $\rightarrow$  (P, R, S, T)

$$(A) \sqrt{\frac{(s-a)(s-c)}{\frac{ac}{(s-b)(s-c)(s-a)(s-b)}}} = \frac{b}{s-b} = \frac{2b}{a+c-b} = 2$$

as a, b, c are in A.P.

$$(B) b \sin A = 2bc - (b^2 + c^2 - a^2)$$

$$\sin A = \frac{1}{bc} [2bc - 2bc \cos A] = 2(1 - \cos A)$$

$$= 4 \sin^2 \frac{A}{2} \Rightarrow \cot \frac{A}{2} = 2$$

$$\sin A = \frac{2 \cdot \frac{1}{2}}{1 + \frac{1}{4}} = \frac{4}{5}, \cos A = \frac{3}{5}$$

$$\text{so } x = 5 \sin A, 5 \cos A \text{ or } x = 3, 4$$

$$(C) AD = \frac{1}{2} \sqrt{2 \left( \frac{131}{2} + 9 \right) - 49}$$

$$= \frac{1}{2} \sqrt{100} = 5$$

$$AB, DE = AB \cdot DC$$

$$\Rightarrow DE = \frac{7 \cdot 7}{5} = \frac{49}{20}$$

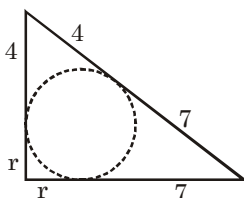
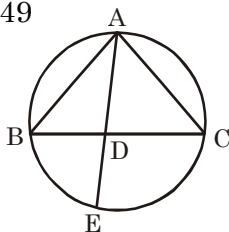
$$[AE] = \left[ 5 + \frac{49}{20} \right] = 7$$

$$(D) \sum \tan \frac{A}{2} \tan \frac{B}{2} = 1$$

$$\Rightarrow \frac{r}{7} \cdot \frac{r}{4} + \frac{r}{7} \cdot 1 + \frac{r}{4} \cdot 1 = 1$$

$$r^2 + 11r = 28$$

$$\Delta = r(11 + r) = 28$$



### SECTION - IV

1. Ans. 6

$$x + 1 = y$$

$$(y + 2)^2 (y + 3)^3 (x + 4)^4$$

$$= y^9 + a_1 y^8 + a_2 y^6 + \dots + a_8 y + a_9$$

$$a_2 = \frac{1}{2} [(2 + 2 + 3 + 3 + 3 + 4 + 4 + 4 + 4)^2 - (2 \cdot 2^2 + 3 \cdot 3^2 + 4 \cdot 4^2)]$$

$$= \frac{1}{2} [29^2 - 99] = 371$$

2. Ans. 3

$|\vec{\Delta} \cdot \hat{n}|$  is minimum area.

$$\vec{\Delta} = \vec{AB} \times \vec{AC} \quad A(3, 3, 3)$$

$$= (-3\hat{i}) \times (-3\hat{i} - 3\hat{k}) \quad B(3, 3, 0)$$

$$= -9\hat{j} \quad C(0, 3, 0)$$

$$\hat{n} = \frac{1}{3} (2\hat{i} + \hat{j} + 2\hat{k}) \quad D(0, 3, 3)$$

$$|\vec{\Delta} \cdot \hat{n}| = 3$$

3. Ans. 9

$$(1 + x^5)^{20} + {}^{20}C_1 x^7 (1 + x^5)^{19} + {}^{20}C_2 x^{14} (1 + x^5)^{18} + \dots$$

$$20 \cdot {}^{19}C_2 = 20 \cdot 19 \cdot 9$$

4. Ans. 7

$$x^3 + y^3 = 8\lambda$$

(I) x, y both even  $\rightarrow$  all favourable

$${}^{50}C_2 = \frac{50 \times 49}{2}$$

(II) x, y both odd

category

Number of numbers

$$(i) 8\lambda + 1 \rightarrow 1, 9, \dots, 97 \rightarrow 13$$

$$(ii) 8\lambda + 3 \rightarrow 3, 11, \dots, 99 \rightarrow 13$$

$$(iii) 8\lambda + 5 \rightarrow 5, 13, \dots, 93 \rightarrow 12$$

$$(iv) 8\lambda + 7 \rightarrow 7, 15, \dots, 91 \rightarrow 12$$

$$x \in (i) \quad y \in (iv) \rightarrow (13 \times 12)$$

$$x \in (ii) \quad y \in (iii) \rightarrow (13 \times 12)$$

$$\text{Ans. } 25 \times 49 + 312 = 1537$$

5. Ans. 7

$$p = \left(\frac{1}{6}\right)^3 \frac{1}{6} + 3 \left(\frac{1}{6}\right)^3 + 3 \left(\frac{1}{6}\right)^3 \frac{1}{6} + 6 \left(\frac{1}{6}\right)^3 \frac{1}{6} + 3 \left(\frac{1}{6}\right)^3 \frac{1}{6} + 9 \left(\frac{1}{6}\right)^3 \frac{1}{6} = \frac{25}{6^4}$$

# JEE (Main + Advanced) : LEADER COURSE

## PHASE- III to VII (SCORE-I)

Test Type : PART TEST

Test Pattern : JEE-Main

**TEST DATE : 05 - 03 - 2017**

### PAPER-2

### SOLUTION

1. **Ans. (3)**

**Sol.** Out-put is XOR GATE

A & B two are input

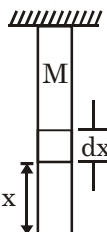
$$\text{XOR out put} = A \cdot \bar{B} + B \cdot \bar{A}$$

2. **Ans. (2)**

**Sol.** Sine magnification is  $-2$  so virtual object have virtual image.

virtual object to virtual image then mirror is convex.

3. **Ans. (2)**



**Sol.**

Tension in rod at a distance  $x$  from lower end is  $\frac{mgx}{l}$

$Y$  is young modulus of elasticity

then change in length in  $dx$  element is  $dy$

$Y \times \text{strain} = \text{stress}$

$$Y \times \frac{dy}{dx} = \frac{T}{A}$$

$$Y \times \frac{dy}{dx} = \frac{mgx}{l \times A}$$

$$\int_0^y Y dy = \int_{30}^{70} \frac{mg}{lA} \times x dx$$

$$Y y = \frac{mg}{lA} \left[ \frac{(70)^2 - (30)^2}{2} \right]$$

$$y = \frac{mg}{AY \times 100} \times 2000$$

$$y = \frac{mg}{AY} \times 20$$

$$\text{Total length is} = 40 \text{ cm} = 20 \frac{mg}{AY} \text{ cm}$$

4. **Ans. (3)**

**Sol.**  $qE = qvB$

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB} = \frac{mE}{qB^2}$$

5. **Ans. (2)**

**Sol.** Some of the characteristics of an optical fibre are as follows

(i) This works on the principle of total internal reflection.

(ii) It consists of core made up of glass/silica/plastic with refractive index  $n_1$ , which is surrounded by a glass or plastic cladding with refractive index  $n_2$  ( $n_2 > n_1$ ). The refractive index of cladding can be either changing abruptly or gradually changing (graded index fibre).

(iii) There is a very little transmission loss through optical fibres.

(iv) There is no interference from stray electric and magnetic field to the signals through optical fibres.

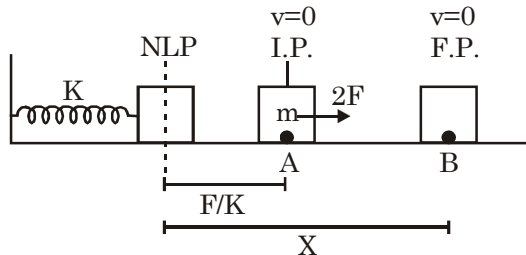
6. **Ans. (3)**

7. **Ans. (1)**

8. **Ans. (2)**

9. Ans. (3)

Sol. Replacing it with string block system



Let at initial position  $2F$  force is applied then W.E.T. from A to B

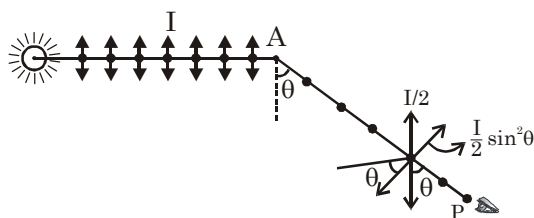
$$W_s + W_F = 0$$

$$X = \frac{3F}{K}$$

$$\text{Net elongation in spring} = \frac{3F}{K} \Rightarrow 3CE$$

10. Ans. (3)

Sol. Eye receive all component of light which is along the line. AP



$$\text{And perpendicular component is } \frac{I}{2} \sin^2 \theta$$

Net intensity recieved by light is

$$\frac{I}{2} + \frac{I}{2} \sin^2 \theta$$

11. Ans. (1)

$$\text{Sol. } I = \frac{B_0^2}{2\mu} \times C$$

$$I = \frac{(100)^2 \times (10^{-9})^2}{2 \times 4\pi \times 10^{-7}} \times 3 \times 10^8$$

$$I = \frac{10^4 \times 10^{-18} \times 3 \times 10^8}{8\pi \times 10^{-7}}$$

$$I = \frac{30}{89} = 1.19426$$

12. Ans. (4)

13. Ans. (1)

$$\text{Sol. } \pi R^2 B_0 = E \times 2\pi r$$

$$E = \frac{\pi R^2}{2\pi r} B_0 = \frac{R^2 B_0}{2r}$$

$$a = \frac{qE}{m} = \frac{q}{m} \frac{R^2 B_0}{2r} = \frac{q B_0 R^2}{2mr}$$

14. Ans. (4)

Sol. V voltage of source drop across resistance is  $V_R$ , inductor is  $V_L$  and capacitor  $V_C$

Applying Krichhof's law

$$\epsilon V = 0$$

$$V_R + V_L = V_C - V = 0$$

15. Ans. (2)

Sol. Temperature of surrounding is  $20^\circ\text{C}$

$$50^\circ\text{C}$$

$$35^\circ\text{C}$$

$$\text{Heater } 100 \text{ W}$$

$$\text{Heater } XW$$

$$100 = K(50 - 20)$$

$$x = K(35 - 20)$$

$$100 = K \times 30$$

$$x = \frac{100}{30} \times 15$$

$$x = 50$$

16. Ans. (2)

Sol. F = pressure at centroid  $\times$  area

$$F = \left( P_0 + \frac{h + h + a + h + a}{3} \rho g \right) \frac{1}{2} ab$$

$$= \left( P_0 + \left( h + \frac{2a}{3} \right) \rho g \right) \frac{1}{2} ab$$

17. Ans. (4)

18. Ans. (1)

Sol. Maximum current thorough capacitor =  $\frac{V_0}{Z}$

$$i = \frac{V_0}{Z}$$

$$Q\omega = \frac{V_0}{Z}$$

$$Q = \frac{V_0}{\omega Z}$$

19. Ans. (1)

Sol. On increaing temperature of semiconductor bond between molecule break so more electron become free to move hence. Conducting infreases and mobility decreases.

20. Ans. (4)

Sol. Charge is conserved. In order to fully convert an electron into energy, a positron (the electron's antiparticle must be involved). That is, electron + positron  $\rightarrow$  energy, NOT electron  $\rightarrow$  positron + energy.]

21. Ans. (1)

**22. Ans. (3)**

| Sol. | 1 <sup>st</sup> | 2 <sup>nd</sup> | 3 <sup>rd</sup> | 4 <sup>th</sup> |
|------|-----------------|-----------------|-----------------|-----------------|
| V.S. | 0.95            | 0.90            | 0.85            | 0.80            |
| GAP  | 0.05            | 0.10            | 0.15            | 0.20            |

difference is = 0.5 mm = 0.05 cm

Net reading is = 3.1 + 0.05 = 3.15 cm

**23. Ans. (2)**
**Sol.** Index error in u = +1cm

u = 8 cm

index error in v = -1cm

v = 17 + 1 = 18 cm

$$\frac{1}{f} = \frac{1}{18} + \frac{1}{8}$$

f = 5.53 cm

**24. Ans. (1)**
**25. Ans. (3)**
**26. Ans. (4)**
**Sol.** Damping coefficient =  $2\sqrt{km}$ 

$$= 2m\omega$$

$$\propto m$$

$$m_A = m \quad m_B = \frac{m}{2}$$

$$\frac{b_A}{b_B} = \frac{1}{2}$$

**27. Ans. (3)**
**Sol.**  $PV = \frac{m}{M}RT$ 

$$\frac{PVM}{Rm} = T \quad T \propto M \therefore T_{He} < T_{N_2}$$

**28. Ans. (2)**
**29. Ans. (2)**
**30. Ans. (3)**
**Sol.**  $\lambda_{\min}(p = \infty)\lambda_{\min} = 1500\text{\AA}$ ;

$$\Delta E = -E_1 = \frac{12420}{1500} \text{ eV} = 8.28 \text{ eV}$$

Hence ionization potential is 8.28V

**31. Ans. (4)**
**32. Ans. (4)**
**33. Ans. (2)**
**34. Ans. (4)**
**35. Ans. (4)**

$$\text{Collision frequency } Z_{11} = \frac{1}{2}(\pi\sigma^2)$$

$$(\sqrt{2}\mu_{\text{Avg.}})(N^o)^2$$

$$\left[ N^o = \frac{P}{KT} \right]$$

$$Z_{11} = \frac{1}{2}(\pi\sigma^2)\sqrt{2}\sqrt{\frac{8RT}{\pi M}}\left(\frac{P}{KT}\right)^2$$

$$Z_{11} \propto \frac{1}{T^{3/2}}$$

**36. Ans. (2)**
**37. Ans. (2)**

$$P_0 = 76$$

$$P_s = 38$$

$$\frac{P_0 - P_s}{P_s} = 1 = \frac{i(1)}{(2)}$$

$$i = 2 = 1 + (2) \propto$$

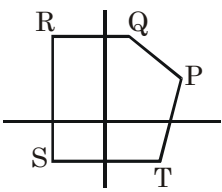
$$2 \propto 1$$

$$\propto = \frac{1}{2}$$

**38. Ans. (4)**
**39. Ans. (4)**
**40. Ans. (3)**
**41. Ans. (2)**
**42. Ans. (1)**
**43. Ans. (3)**
**44. Ans. (4)**
**45. Ans. (1)**
**46. Ans. (3)**
**47. Ans. (4)**
**48. Ans. (2)**
**49. Ans. (3)**
**50. Ans. (1)**
**51. Ans. (3)**
**52. Ans. (1)**
**53. Ans. (4)**
**54. Ans. (3)**
**55. Ans. (1)**
**56. Ans. (1)**
**57. Ans. (2)**
**58. Ans. (3)**
**59. Ans. (4)**
**60. Ans. (2)**

61. Ans. (4)

$P(\alpha, \beta), Q(\beta, \alpha), R(-\beta, \alpha), S(-\beta, -\alpha), T(\beta, -\alpha)$



$$187 = 4\alpha\beta + \frac{1}{2} \cdot 2\alpha(\alpha - \beta)$$

$$187 = \alpha(\alpha + 3\beta)$$

$$\alpha = 11, 3\beta + \alpha = 17$$

$$\alpha = 11 \text{ and } \beta = 2$$

62. Ans. (1)

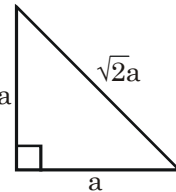
$$x^2 - \sqrt{2}x + 1 = 0$$

$$\therefore \alpha = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}, \beta = \frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}$$

$$= e^{i\pi/4} = e^{-i\pi/4}$$

$$\alpha^{50} + \beta^{50} = e^{i25\pi/2} + e^{-i25\pi/2} = i + (-i) = 0$$

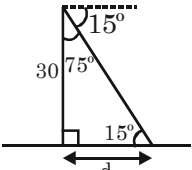
63. Ans. (1)

$$r = \frac{\frac{1}{2}a^2}{a + \frac{a}{\sqrt{2}}} \Rightarrow 1 = \frac{a}{2 + \sqrt{2}}$$


$$\Delta = \frac{1}{2}a^2 = \frac{1}{2}(4 + 2 + 4\sqrt{2})$$

$$= 3 + 2\sqrt{2}$$

64. Ans. (3)



$$\tan 15^\circ = \frac{30}{d}$$

$$d = \frac{30}{2 - \sqrt{3}} = \frac{30(\sqrt{3} + 1)}{(\sqrt{3} - 1)}$$

65. Ans. (3)

It is always true for  $n \geq 5$

66. Ans. (1)

$P_1$  and  $P_2$  are  $x + 2y - 2z = 0$   
and  $2x - 3y + 6z = 0$

$$\cos \alpha = \frac{|2 - 6 - 12|}{3.7} = \frac{16}{21}$$

67. Ans. (4)

$$|\text{adj } 3P| = |3P|^3 = 3^{12} |P|^3 = -3^{12} \cdot 2^3$$

68. Ans. (2)

Given conic C is parabola  
focus : (1, -1)  
E :  $3x - 4y = 0$

minimum distance is  $\perp$  distance

$$\Rightarrow \text{distance} = \left| \frac{6 + 4}{5} \right| = 2$$

69. Ans. (1)

$P(A) = 0.3$   
 $P(A \cup B) = P(A) + P(B) - P(A)P(B)$   
 $\Rightarrow 0.8 = 0.3 + P(B) - 0.3 P(B)$

$$\Rightarrow P(B) = \frac{5}{7}$$

$$P(A \rightarrow B) = 1 - P(A \cap \bar{B})$$

$$= 1 - (P(A) - P(A)P(B))$$

$$= 1 - \left( 0.3 \times \frac{2}{7} \right) = \frac{32}{35}$$

70. Ans. (1)

$${}^nC_{r-2} = 36, {}^nC_{r-1} = 84, {}^nC_r = 126$$

$$\frac{{}^nC_{r-1}}{{}^nC_{r-2}} = \frac{84}{36} \Rightarrow \frac{n-r+2}{r-1} = \frac{7}{3}$$

$$\Rightarrow 3n + 13 = 10r \quad \dots(1)$$

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{126}{84} \Rightarrow \frac{n-r+1}{r} = \frac{3}{2}$$

$$\Rightarrow 2n + 2 = 5r \quad \dots(2)$$

$$\therefore n = 9, r = 4$$

$${}^nC_{2r} = {}^9C_8 = 9$$

71. Ans. (1)

Let  $|\vec{U}| = 2 \cos \hat{\alpha} \hat{i} + 2 \sin \hat{\alpha} \hat{j}$

$$[(\vec{U} \cdot \vec{V} \cdot \vec{W})]^2 = \begin{vmatrix} 2 \cos \alpha & 2 \sin \alpha & 0 \\ 2 & 1 & -1 \\ 1 & 0 & 3 \end{vmatrix}^2$$

$$= |6 \cos \alpha - 14 \sin \alpha|^2$$

Maximum value =  $36 + 196 = 232$

72. Ans. (1)

$$\sin \alpha = \frac{1}{3}$$

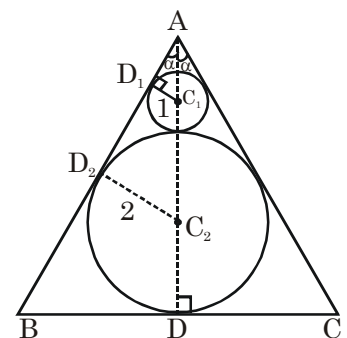
$$\therefore AC_1 = 3$$

$$AC_2 = 6$$

$$AD = 8$$

$$\therefore BD = 2\sqrt{2}$$

$$\text{Area} = \frac{1}{2} \cdot 4\sqrt{2} \cdot 8 = 16\sqrt{2}$$



73. Ans. (1)

$$|z - 3 - 4i| = 4$$

$$||z| - 5| \leq 4$$

$$|z| \leq 9$$

74. Ans. (3)

$$\begin{vmatrix} 12 & b & c \\ a & 24 & c \\ a & b & 36 \end{vmatrix} = 0$$

$$(12)(24)(36) - 12bc - 36ba + 2abc - 24ac = 0$$

....(i)

$$\frac{1}{a-12} + \frac{2}{b-24} + \frac{3}{c-36}$$

$$= \frac{(b-24)(c-36) + 2(a-12)(c-36) + 3(a-12)(b-24)}{(a-12)(b-24)(c-36)}$$

on solve

$$= -\frac{1}{6}$$

75. Ans. (4)

$$6 \times 5!$$

76. Ans. (3)

$y = mx + 1$  is tangent to ellipse

$$x^2 + 4y^2 = 1 \text{ in Ist quadrant } \therefore m < 0$$

$$\therefore 1 = m^2 + \frac{1}{4}$$

$$m = \frac{\sqrt{3}}{2} \text{ or } -\frac{\sqrt{3}}{2}$$

(reject)

77. Ans. (2)

Use : contrapositive of  $p \rightarrow q$  is  $(\sim q) \rightarrow (\sim p)$

78. Ans. (2)

(1,2,3) lie on

$$L \therefore -\frac{1}{2} = \frac{1}{b} = \frac{4}{c}$$

$$\therefore b = -2, c = -8$$

Line L and k are parallel

$$\frac{2}{a} = \frac{b}{2} = \frac{c}{d} \Rightarrow \frac{2}{a} = -1 = -\frac{8}{d}$$

$$\therefore a = -2, d = 8$$

$$L : \frac{x-2}{2} = \frac{y-1}{-2} = \frac{z+1}{-8}$$

$$K : \frac{x+2}{-2} = \frac{y-3}{2} = \frac{z+4}{8}$$

$$\cos \theta = \frac{|8+4-24|}{\sqrt{72}\sqrt{29}} = \frac{\sqrt{2}}{\sqrt{29}}$$

$$\therefore \sin \theta = \frac{\sqrt{27}}{\sqrt{29}}$$

79. Ans. (3)

$$\sum_{n=0}^4 (1009 - 2n)^4 {}^4C_n (-1)^n$$

$$(1009)^4 - 4(1007)^2 + 6(1005)^4 - 4(1003)^4 + (1001)^4$$

$$(1005 + 4)^4 + (1005 - 4)^4$$

$$-4[(1005 + 2)^4 + (1005 - 2)^4] + 6(1005)^4$$

$$= 512 - 4 \times 32 = 384$$

80. Ans. (2)

$$2p - 3q + 12r = 5$$

$$b = p^2 + q^2 + r^2$$

$$c = pq - qp + qr - qr + 3r^2 = 3r^2$$

$$b + c = p^2 + q^2 + 4r^2$$

$$\text{use : } (2\hat{i} - 3\hat{j} + 6\hat{k}) \cdot (p\hat{i} - q\hat{j} + 2r\hat{k}) \leq \sqrt{2^2 + 3^2 + 6^2} \sqrt{p^2 + q^2 + 4r^2}$$

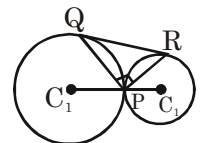
$$\frac{25}{49} \leq p^2 + q^2 + 4r^2$$

81. Ans. (3)

$$PQ^2 + PR^2 + QR^2 = 2QR^2$$

$$= 2((C_1 C_2)^2 - (r_1 - r_2)^2)$$

$$= 2[36 - 4] = 64$$



82. Ans. (2)

$$a R b \Leftrightarrow a = 2^k a \text{ it is true for } k = 0$$

$\therefore$  reflexive

$(2,1) \in R$  but  $(1,2) \notin R \Rightarrow$  it is not symmetric

if  $a = 2^{k_1} b$  and  $b = 2^{k_2} c$ , then  $a = 2^{k_1+k_2} c$

$\Rightarrow$  it is transitive.

83. Ans. (4)

$$P = \frac{2 \times 19}{10 \times 9} = \frac{19}{45}$$

favourable :  $\{(3,4), (3,5), \dots, (3,10)$

$(6,7), (6,8), \dots, (6,10)$

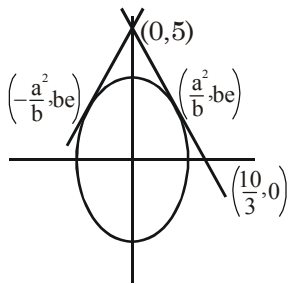
$(9,10),$

$(1,4), (2,4)$

$(1,8), (2,8), (4,8), (5,8), (7,8)\}$

84. Ans. (4)

Let ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



tangent  $\frac{x}{b} + \frac{ye}{b} = 1$

$$\therefore \frac{b}{e} = 5$$

$$\text{and } b = \frac{10}{3} \Rightarrow e = \frac{2}{3}$$

$$\Rightarrow a^2 = \frac{500}{81} \therefore L.R = \frac{2a^2}{b} = \frac{100}{27}$$

85. Ans. (2)

Determinant value of matrix

$$= 1 - wc - aw + w^2ac = 0$$

$$\Rightarrow (1 - aw)(1 - wc) = 0$$

$a = \frac{1}{w} = w^4 \Rightarrow b$  and  $c$  each have 4 and 4 options.

if  $c = \frac{1}{w} = w^4$  and  $a \neq w^4$

$\Rightarrow a$  have 3 and  $b$  have 4 options.

$$\therefore \text{Total matrices} = 4 \times 4 + 3 \times 4 = 28$$

86. Ans. (3)

Mean of  $\omega_i = \ell(\text{mean of } y_i) + k$

$$55 = \ell.48 + k \quad \dots(i)$$

standard deviation of

$$\omega_i = \ell(\text{standard deviation of } y_i)$$

$$15 = \ell.12 \quad \dots(ii)$$

$$\ell = 1.25 \text{ and } k = -5$$

87. Ans. (2)

If circle intersect at 4 points then

sum of x coor = 0

$$\therefore \text{Points } (17, 289), (-2, 4), (13, 169), (-28, 784)$$

$$\text{directrix is } y = -\frac{1}{4}$$

sum of perpendicular distances

$$= \left(289 + \frac{1}{4}\right) + \left(4 + \frac{1}{4}\right) + \left(169 + \frac{1}{4}\right) + \left(784 + \frac{1}{4}\right) = 1247$$

88. Ans. (4)

$$|\vec{B}|^2 = A_x^2 + A_y^2 + A_z^2$$

$$= (\text{Area of } \Delta PQR)^2$$

$$= \left( \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & -1 & -1 \\ 2 & 2 & 1 \end{vmatrix} \right)^2 = \frac{1+1+16}{4}$$

$$= \frac{9}{2}$$

89. Ans. (4)

$$np = 2$$

$$npq = 1 \therefore p = q = \frac{1}{2}, n = 4$$

$$P(x) = 1 - {}^4C_0 \left(\frac{1}{2}\right)^4 - {}^4C_1 \left(\frac{1}{2}\right)^4 = 1 - \frac{1}{16} - \frac{4}{16} = \frac{11}{16}$$

90. Ans. (3)

$$4x^3 + 4y^3 = xy(xy + 16)$$

$$4x^3 + 4y^3 - x^2y^2 - 16xy = 0$$

$$(4x - y^2)(x^2 - 4y) = 0$$

$\therefore$  it is combined equation of two parabola

$\Rightarrow$  given tangent is common tangent of parabola  $y^2 = 4x$  and  $x^2 = 4y$

$\therefore$  given tangent is  $x + y + 1 = 0$

$$\Rightarrow \alpha = \beta = 1$$